

CSI132 - DISCRETE STRUCTURES II TUTORIAL
ATTEMPT ALL QUESTIONS
DURATION: 2 HOURS
PERMUTATIONS, COMBINATIONS, BINOMIAL COEFFICIENTS

1. How many permutations are there of the set $\{a, b, c, d, e, f, g\}$?
2. Let $S = \{1, 2, 3, 4, 5\}$
 - (a) List all the three permutations of set S .
 - (b) List all the three combination of set S .
3. How many possibilities are there for a *win, place, and show* (first, second, third) positions in a horse race with 12 horses if all orders of finish are possible?
4. A group contains n men and n women. How many ways are there to arrange these people in a row if the men and women alternate.
5. How many subsets with an odd number of elements does a set with 10 elements have?
6. A coin is flipped 10 times where each flip comes up either tails or heads. How many possible outcomes:
 - (a) are there in total?
 - (b) contain exactly two heads?
 - (c) contain at most three tails?
 - (d) contain the same number of heads and tails?
7. How many ways are there for eight men and five women to stand in a line so that no two women stand next to each other? [*Hint: First position the men, then consider possible positions for the women.*]
8. Thirteen people on a softball team show up for a game.
 - (a) How many ways are there to choose 10 players to take the field?

- (b) How many ways are there to assign 10 positions by selecting players from the 13 people who showed up?
 - (c) Of the thirteen people who showed up, three are women. How many ways are there to choose 10 players to take the field if at least one of the players must be a women?
- 9. A professor writes 40 discrete structures true/false questions. Of the statements in these questions, 17 are true. If the questions can be positioned in any order, how many different answer keys are possible?
- 10. 7 women and 9 men are on the faculty in the mathematics department at a University.
 - (a) How many ways are there to a committee of 5 members of the department if at least 1 woman must be in the committee?
 - (b) How many ways are there to a committee of 5 members of the department if at least 1 woman and 1 man must be in the committee?
- 11. How many strings of 6 lowercase letters from the English alphabet contain:
 - (a) the letter a ?
 - (b) the letters a and b ?
 - (c) The letters a and b in consecutive position with a preceding b , with all letters distinct?
 - (d) The letters a and b , where a is somewhere to the left of b in the string, with all letters distinct?
- 12. Suppose that a department contains 10 men and 15 women. How many ways are there to form a committee with 6 members if it must have more women than men.
- 13. Using the binomial theorem, find the expansion of $(x + y)^5$
- 14. Find the coefficient of x^5y^8 in $(x + y)^{13}$
- 15. What is the coefficient of $x^{101}y^{99}$ in the expansion of $(2x - 3y)^{200}$